



University of Milan
Data Science for Economics

Analyzing Thematic Alignment in Scientific Journals (case study: Journal of Machine Learning Research (JMLR))

Text Mining & Sentiment Analysis Project

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1 Introduction

Scientific journals define their identity through their stated Aims & Scope. This text describes the kinds of topics, methods, and contributions that the journal intends to publish. However, research fields are not static. As new methods, applications, and interdisciplinary directions emerge, the actual content of a journal may gradually expand or shift. This creates an important question: to what extent do the articles published in a journal remain thematically aligned with its declared editorial focus?

This project investigates that question using the Journal of Machine Learning Research (JMLR) as a case study. JMLR is a broad and influential journal in machine learning, making it an appropriate target for studying thematic coherence in a fast-moving research field. Rather than relying on manual inspection alone, this project develops a computational approach for comparing the journal’s Aims & Scope with the abstracts of published articles. The central idea is to treat the Aims & Scope as a reference representation of the journal’s intended focus and then measure how closely each article abstract aligns with it.

The project builds on recent developments in natural language processing, especially text embeddings and topic modeling. Traditional lexical methods such as TF-IDF represent documents based on word overlap, which can be useful as a baseline but often fails to capture deeper semantic similarity. Sentence-BERT addresses this limitation by producing sentence-level embeddings that can be efficiently compared using cosine similarity, making it suitable for semantic similarity tasks such as comparing abstracts with scope statements. In addition, BERTopic provides a way to discover latent topics by combining transformer-based document embeddings with clustering and class-based TF-IDF topic representations. UMAP is also used as a dimensionality reduction method for visualizing high-dimensional embeddings in a lower-dimensional space.

Using these methods, the project analyzes a corpus of 467 JMLR article abstracts from 2019 to 2026. Each abstract is compared with the JMLR Aims & Scope using cosine similarity over both TF-IDF vectors and SBERT embeddings, with the SBERT score used as the main semantic alignment measure. The analysis then examines the distribution of alignment scores, identifies high- and low-alignment outliers, investigates topic-level patterns, and tests whether there is evidence of thematic drift over time.

The results suggest that JMLR maintains a broadly stable thematic relationship with its stated scope. The average SBERT alignment score indicates moderate semantic alignment across the corpus, while the temporal trend is almost flat, showing no strong evidence of major thematic drift. At the same time, topic modeling and outlier analysis reveal meaningful variation within the journal’s published content, highlighting the usefulness of embedding-based methods for studying thematic coherence and conceptual evolution in academic publishing.

2 Research Question and Methodology

2.1 Research Objective

The objective of this project is to quantitatively evaluate the thematic alignment between the articles published in the Journal of Machine Learning Research (JMLR) and the journal’s stated Aims & Scope. In academic publishing, a journal’s Aims & Scope defines its intended intellectual territory: the types of problems, methods, and research contributions that the journal aims to publish. However, as a research field evolves, the actual content of published papers may gradually shift, expand, or occasionally diverge from this stated scope. This project addresses that issue by treating the Aims & Scope statement as a reference representation of the journal’s intended focus and comparing it with the semantic content of published article abstracts.

The selected case study is JMLR, a major journal in the field of machine learning. A corpus of article abstracts published between 2019 and 2026 was collected and cleaned for analysis. The main alignment measure is based on sentence-transformer embeddings, while TF-IDF is also used as a lexical baseline. Beyond paper-level alignment, the project also investigates topic-level patterns

and temporal trends in order to provide a more detailed view of how the journal’s published content relates to its stated editorial scope.

Overall, the goal is not to judge whether individual papers “belong” or “do not belong” to the journal in an editorial sense, but to provide an interpretable quantitative framework for studying thematic coherence, outlier publications, and long-term conceptual evolution in scientific publishing.

2.2 Research Question

The main research question addressed in this project is:

How closely do articles published in the Journal of Machine Learning Research align with the journal’s stated Aims & Scope, and does this alignment reveal evidence of thematic drift or outlier publications over time?

This question is motivated by the idea that a scientific journal’s published content should broadly reflect its editorial mission, while still allowing the field to evolve. In the case of JMLR, the journal covers a wide range of machine learning research, including theoretical, methodological, and applied contributions. Therefore, the project does not assume that all papers should be semantically identical to the Aims & Scope statement. Instead, it investigates whether the corpus as a whole remains thematically coherent with the journal’s stated focus.

To answer the main research question, the project is divided into the following sub-questions:

- What is the overall distribution of thematic alignment scores between JMLR article abstracts and the journal’s Aims & Scope?
- Which papers show unusually high or low alignment with the journal scope, and can these cases be interpreted as meaningful thematic matches, outliers, or possible data collection artifacts?
- Does the average thematic alignment score change over the publication period from 2019 to 2026, suggesting possible thematic drift?
- Which latent topics within the corpus are most strongly or weakly aligned with the journal’s stated scope?
- Do topic-level and embedding-space visualizations support the numerical alignment results?

2.3 Data Collection

Paper metadata was retrieved from arXiv using the arxiv Python library. The query jo:”Journal of Machine Learning Research” was used to target papers associated with JMLR, retrieving the 500 most recently submitted papers sorted by submission date in descending order. For each paper, the following fields were collected: title, abstract, author list, publication year, DOI, and arXiv identifier. The resulting corpus spans the years 2019 to 2026.

It is important to note a key limitation of this data collection strategy: the arXiv query does not guarantee that all retrieved papers were formally published in JMLR. arXiv cross-listings from authors who submit to multiple venues may be included, and conversely, JMLR papers not deposited on arXiv would be missed. This has direct implications for the outlier analysis, discussed in Section 3.5.

2.4 Text Preprocessing

Raw abstracts were processed through a lightweight cleaning pipeline. Each abstract was lowercased, whitespace was collapsed, and all characters outside the set of lowercase letters, digits, spaces, hyphens, commas, and full stops were removed using regular expressions. Records with missing or empty abstracts were discarded.

An additional filter was applied to remove papers where the token `jo` appeared in the cleaned abstract. This was necessary because the arXiv query string itself (`jo:"Journal of Machine Learning Research"`) occasionally leaked into metadata fields and introduced spurious matches unrelated to the paper’s content.

The reference document used for alignment scoring is JMLR’s official Aims & Scope text, which describes the journal’s focus on principled machine learning algorithms, statistical learning theory, neural networks, kernel methods, Bayesian approaches, reinforcement learning, representation learning, and the fairness and robustness of ML systems. This text was subjected to the same cleaning pipeline as the abstracts.

JMLR’s Official Aims & Scope - The Reference Document

The Journal of Machine Learning Research (JMLR) provides an international forum for the electronic and paper publication of high-quality scholarly articles in all areas of machine learning. All published papers are freely available online. JMLR has a commitment to rigorous yet rapid reviewing.

JMLR seeks previously unpublished papers on machine learning that contain:

- Novel principled machine learning algorithms and their applications
- Analysis of algorithms and the computational complexity of learning
- Studies on the statistical aspects of learning, including consistency and rates of convergence
- Papers that bridge machine learning and statistics, computational complexity, or other fields
- Experimental and theoretical work on neural networks, kernel methods, Bayesian approaches
- Reinforcement learning, unsupervised learning, supervised learning
- Representation learning, deep learning, optimization for machine learning
- Fairness, robustness, and interpretability in machine learning systems
- Applications of machine learning to real-world problems

2.5 Embedding Models

Two embedding strategies were employed, serving complementary roles.

- **TF-IDF (baseline).** A term frequency-inverse document frequency vectorizer was fitted jointly on the Aims & Scope text and all 500 abstracts, producing a vocabulary of up to 5,000 features with unigrams and bigrams and standard English stopwords removal. TF-IDF captures lexical overlap between the scope text and each abstract but is entirely insensitive to synonymy or semantic proximity; two texts using different words to describe the same concept will receive a low similarity score regardless.
- **Sentence-BERT (SBERT).** The pre-trained all-MiniLM-L6-v2 model from the sentence-transformers library was used to encode all texts into dense 384-dimensional vectors. SBERT produces embeddings in a shared semantic space, meaning that papers discussing concepts related to those in the Aims & Scope — even without direct lexical overlap — will receive higher similarity scores. This is the primary embedding used throughout the analysis; TF-IDF serves as a lexical baseline for comparison.

2.6 Alignment Scoring

Thematic alignment between a paper and the journal’s scope was operationalized as the cosine similarity between their respective embedding vectors. Formally, for a scope vector $\mathbf{s} \in \mathbb{R}$ and an abstract vector $\mathbf{a}_i \in \mathbb{R}$, the alignment score of paper i is defined as:

$$\text{align}(i) = \frac{\mathbf{s} \cdot \mathbf{a}_i}{\|\mathbf{s}\| \cdot \|\mathbf{a}_i\|}$$

Cosine similarity was chosen over Euclidean distance because it is invariant to vector magnitude, which is particularly important for TF-IDF representations where document length would otherwise dominate the similarity measure. Scores range from -1 to 1 , where 1 indicates identical direction in the embedding space, 0 indicates orthogonality, and negative values indicate semantic opposition. This scoring was computed independently for both the TF-IDF and SBERT embeddings.

2.7 Outlier Detection

Papers were flagged as alignment outliers if their SBERT score deviated by more than two standard deviations from the corpus mean. Denoting the mean alignment score as μ and the standard deviation as σ , a paper was labelled a "high outlier" if $align(i) > \mu + 2\sigma$, and a "low outlier" if $align(i) < \mu - 2\sigma$. The sensitivity of this rule was evaluated at three threshold levels (1.5σ , 2.0σ , and 2.5σ) to verify that findings were not artifacts of the chosen cutoff.

2.8 Topic modeling

To characterise the thematic structure of the corpus beyond the single alignment score, topic modeling was performed using BERTopic. BERTopic combines HDBSCAN density-based clustering on precomputed SBERT embeddings with a class-based TF-IDF transformer (c-TF-IDF) to extract interpretable keyword labels for each discovered cluster. The model was configured with a target of 20 topics and a minimum cluster size of 8 papers, with a bigram-capable `CountVectorizer` and `reduce_frequent_words=True` to suppress uninformative high-frequency terms. Each topic was labelled by its top three c-TF-IDF keywords. Papers that HDBSCAN could not assign to any cluster were placed in a noise class (topic -1) and excluded from topic-level summaries, though they are retained in all other analyses.

2.9 Dimensionality Reduction and Visualization

For exploratory visualization, SBERT embeddings were projected to two dimensions using UMAP (Uniform Manifold Approximation and Projection). UMAP was configured with `n_neighbors=15`, `min_dist=0.1`, cosine distance metric, and a fixed random seed for reproducibility. The resulting 2D coordinates were used to produce three complementary views of the corpus: colored by topic assignment, by SBERT alignment score, and by publication year. These views allow the geometric structure of the semantic space to be inspected directly and cross-referenced with the quantitative findings from alignment scoring and topic modeling.

3 Experimental Results

3.1 Dataset Overview

The corpus initially consisted of 500 papers retrieved from arXiv. After preprocessing (dropping records with missing or empty abstracts and filtering entries where the query token `jo` appeared in the cleaned text) 467 papers were retained, spanning the years 2019 to 2026. Table 1 summarizes the year-by-year distribution as visible from the temporal drift plot.

The distribution is notably uneven. The corpus peaks in 2021 with 113 papers, followed by a sharp decline through 2022–2024 and a near-total drop-off in 2025 and 2026. This pattern almost certainly reflects the arXiv retrieval strategy. The query fetched the 500 most recently submitted papers sorted by submission date, but the `jo`: journal field is populated when a paper is indexed against a journal, which can lag publication by months or years. As a result, very recent papers (2025–2026) are underrepresented not because JMLR published fewer papers in those years, but because the journal metadata had not yet been updated on the arXiv at the time of collection. This sampling imbalance should be kept in mind when interpreting temporal trends, particularly

Table 1: Number of papers per year after preprocessing

Year	Papers
2019	34
2020	94
2021	113
2022	77
2023	80
2024	45
2025	20
2026	4
Total	467

the topic drift results in Section 3.7 where 2026 shares are based on only 4 papers and are therefore not reliable.

3.2 Alignment Score Distributions

Figure 1 shows the distribution of alignment scores produced by both embedding methods across the 467 papers. The exact corpus statistics are: SBERT mean = 0.3269, std = 0.0909; TF-IDF mean = 0.0428, std = 0.0368.

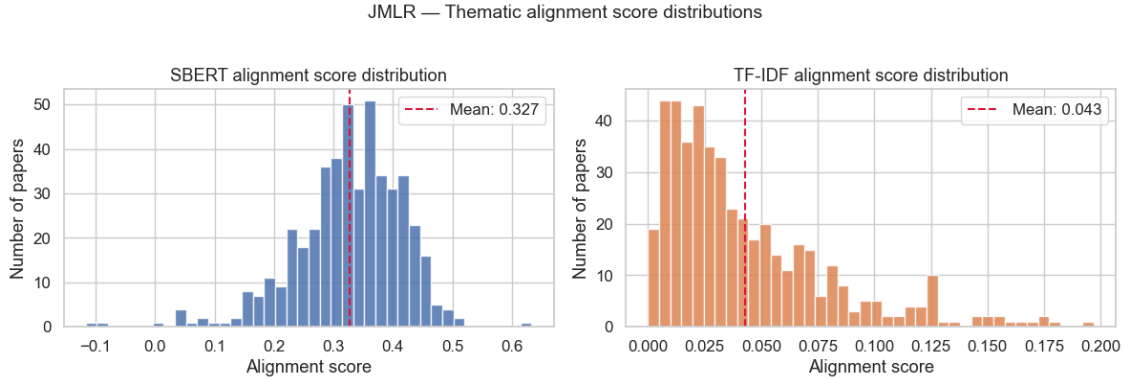


Figure 1: Distribution of SBERT (left) and TF-IDF (right) alignment scores across the corpus.

The two methods show quite different distributions. For SBERT, the distribution is almost bell-shaped and centered around 0.327, with scores going from about -0.12 to 0.65 . This means SBERT is capturing real semantic differences between the papers. Papers that are conceptually close to machine learning get higher scores, while papers that are not really related stay around zero or even below zero. So, the near-normal shape suggests that SBERT works well as a continuous measure of thematic similarity.

On the other hand, the TF-IDF distribution is highly right-skewed. Its mean is only 0.043, and most of the scores are very close to zero. This also makes sense because TF-IDF mainly depends on exact word overlap. Since the Aims & Scope text has a limited and specific set of words, many abstracts, even relevant ones about machine learning, do not share many exact terms with it. Therefore, TF-IDF is better treated as a conservative lexical baseline. It supports the general direction of the SBERT results, but it should not be considered the main scoring method.

3.3 Temporal Drift Analysis

To investigate whether JMLR’s thematic focus has shifted over time, the mean SBERT alignment score was computed per year and plotted alongside a linear trend line, as shown in Figure 2.

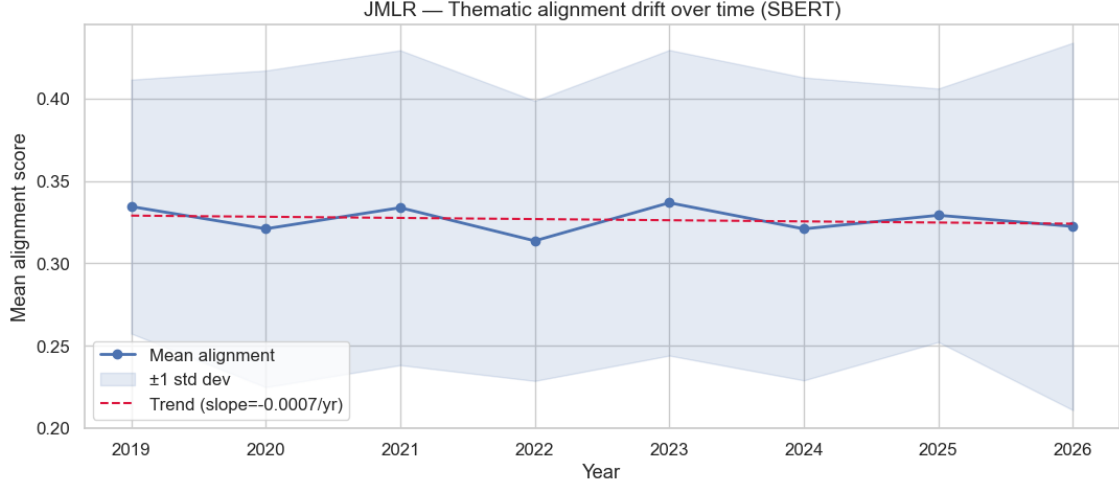


Figure 2: Mean SBERT alignment score per year (2019–2026) with ± 1 standard deviation band and linear trend.

The results show that the yearly average alignment scores remain relatively stable over time. The fitted trend line has a very small negative slope of approximately -0.0007 per year, which suggests only a weak decrease in alignment across the analyzed period. In practical terms, this trend is too small to support a strong claim of thematic drift. Instead, the journal appears to maintain broadly consistent thematic alignment with its stated scope.

This does not mean that all years contain the same types of papers. Some variation is visible across years, and individual articles or topics may still deviate from the general pattern. However, at the corpus level, the temporal analysis does not provide strong evidence that JMLR has moved substantially away from its Aims & Scope during the 2019–2026 period.

3.4 Top and Bottom Papers by Alignment Score

Table 2 and Table 3 report the ten highest- and ten lowest-scoring papers by SBERT alignment score. Figure 3 presents the same information as a horizontal bar chart.

Table 2: Top 10 papers by SBERT alignment score.

Rank	Title	Year	SBERT Score
1	The Bayesian Learning Rule	2021	0.6329
2	A Framework for Improving the Reliability of B...	2022	0.5159
3	Physics-informed kernel learning	2024	0.5042
4	Infeasible Deterministic, Stochastic, and Vari...	2023	0.4916
5	Generic Unsupervised Optimization for a Latent...	2020	0.4914
6	Gaussian Process Inference Using Mini-batch St...	2021	0.4886
7	Reward Shaping for Human Learning via Inverse...	2020	0.4850
8	Recent Trends in Artificial Intelligence Techn...	2023	0.4726
9	Improved Random Features for Dot Product Kernels	2022	0.4726
10	Stochastic interior-point methods for smooth c...	2024	0.4718

The top-ranked papers look very consistent and clearly belong to core machine learning areas, such as Bayesian inference, kernel methods, Gaussian processes, stochastic optimization, and reinforcement learning. Their scores are mostly between 0.47 and 0.63, which shows that the metric is able to detect papers that are thematically close to the journal’s scope. The Bayesian

Table 3: Bottom 10 papers by SBERT alignment score.

Rank	Title	Year	SBERT Score
1	Origin of Jumping Oscillons in an Excitable Re...	2021	-0.1165
2	Irradiation of TiBe12 and CrBe12 with He+ ions...	2023	-0.0940
3	Classifying links and spatial graphs with fini...	2023	0.0020
4	N-quandles of spatial graphs	2022	0.0380
5	The First Photometric Analysis of Two Low Mass...	2024	0.0388
6	Global solutions and Relaxation Limit to the C...	2020	0.0409
7	Histogram lies about distribution shape and Pe...	2021	0.0458
8	CopyCat: Controlled Instruction-Level Attacks...	2020	0.0546
9	Testing Conditional Independence via Quantile...	2020	0.0807
10	Axiomatizing Maximal Progress and Discrete Time	2020	0.0817

Learning Rule is the strongest case, with a score of 0.633. This makes sense because it is strongly aligned with the probabilistic and statistical language used in the Aims & Scope text.

The bottom-ranked papers show the opposite pattern. The two lowest-scoring papers are a physics paper about reaction-diffusion oscillons, with a score of -0.117, and a materials science paper about beryllide irradiation, with a score of -0.094. These negative scores suggest that, in the SBERT embedding space, their semantic content is not just weakly related, but actually far from the journal’s scope. The other low-ranked papers are also from areas such as pure mathematics, astrophysics, graph topology, quandle theory, and formal process algebra. So, these papers are most likely arXiv cross-listing artifacts that matched the journal query filter, but were not actually published in JMLR.

3.5 Outlier Analysis

As the next step, articles with unusually high or low semantic alignment to the JMLR Aims & Scope were identified. Using the SBERT alignment scores, papers were classified as outliers based on a standard deviation threshold around the corpus mean. Figure 4 shows the full alignment score distribution with the 2σ outlier thresholds overlaid. With a corpus mean of 0.3269 and standard deviation of 0.0910, the upper bound is 0.5089 and the lower bound is 0.1449.

At the 2σ threshold, 2 papers are flagged as high outliers and 14 as low outliers. Table 4 reports the sensitivity of these counts to the choice of threshold.

Table 4: Sensitivity of outlier counts to threshold selection.

Threshold	High outliers	Low outliers
1.5σ	12	32
2.0σ	2	14
2.5σ	1	10

The 2σ threshold seems like a reasonable point to work with. It is strict enough to flag papers that are genuinely unusual, but not too strict to miss important outliers. The interesting point is that the outliers are not balanced. There are 14 low outliers but only 2 high outliers. This shows that the main scope problem for JMLR is not that the journal is becoming too focused on one narrow ML subfield. Instead, the bigger issue is contamination in the corpus from papers outside machine learning.

The 14 low outliers can be divided into three clear groups. The first group includes papers with negative scores, such as physics simulations, materials science, and pure mathematics papers. These are semantically very far from machine learning and are most likely arXiv cross-listing artifacts. The second group includes papers with scores close to zero, around 0.00 to 0.05. These include topology and astrophysics papers, and they probably have the same explanation.

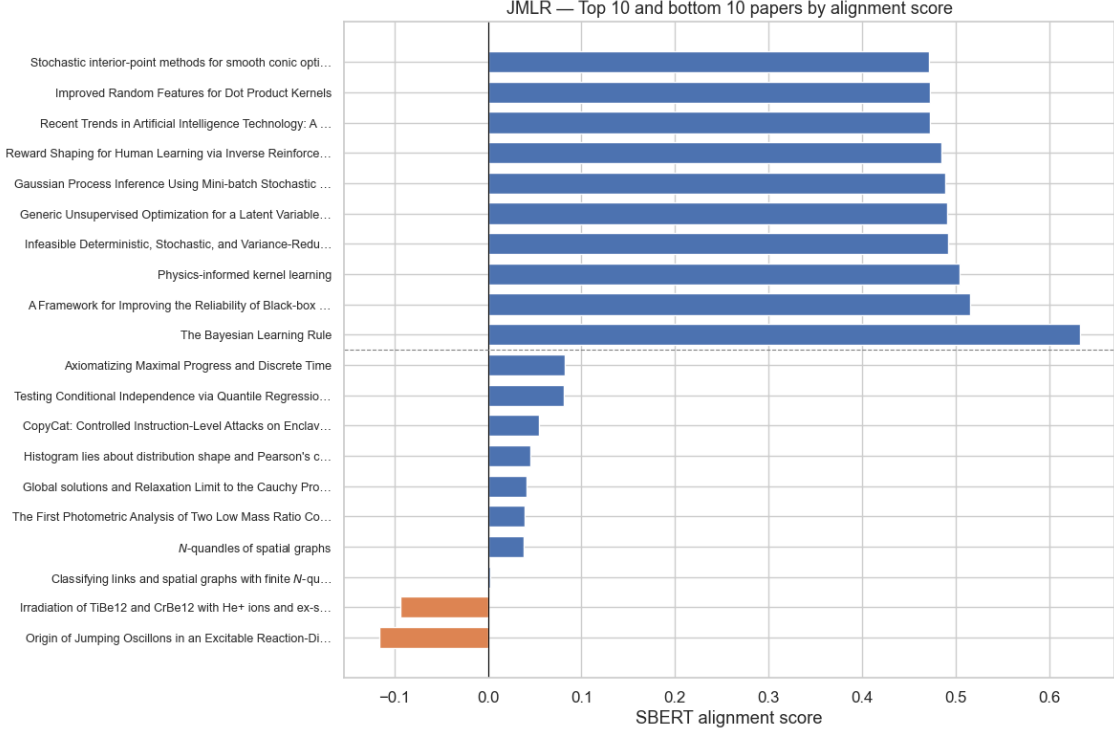


Figure 3: Top 10 and bottom 10 papers by SBERT alignment score.

The third group is more ambiguous, with borderline scores between 0.05 and 0.14. Some of these papers are related to statistics, such as “Histogram lies about distribution shape” and “Testing Conditional Independence via Quantile Regression”. So, they are methodologically connected to machine learning, but their vocabulary is different enough that SBERT places them below the threshold. At the same time, this group also includes papers from unrelated engineering fields, so not all borderline cases should be interpreted in the same way.

3.6 Topic Modeling Results

Beyond individual paper scores, topic modeling was used to examine how broader research themes within the JMLR corpus relate to the journal’s Aims & Scope. BERTopic identified groups of semantically similar abstracts and assigned each paper to a topic. This made it possible to move from paper-level alignment to topic-level interpretation, where the average SBERT alignment score was computed for each discovered topic.

BERTopic identified 13 coherent thematic clusters from the 467 abstracts, with 97 papers (approximately 21%) assigned to the noise class (topic -1) by HDBSCAN. Table 5 summarizes the discovered topics, and Table 6 reports their mean SBERT alignment scores.

All 13 clusters match recognized machine learning subfields, which shows that BERTopic is able to recover the main intellectual structure of the journal. However, the alignment scores also show an important pattern that needs to be interpreted carefully. Topics whose vocabulary directly overlaps with the Aims & Scope text get the highest scores. For example, Bayesian and variational inference scores 0.372, neural networks scores 0.364, loss and classification scores 0.355, and reinforcement learning also scores 0.355.

On the other hand, some topics are still completely valid JMLR topics, but they receive lower scores because they use more specialized or domain-specific vocabulary. This includes causal inference with 0.250, time series and dynamics with 0.253, and graph methods with 0.279. So, this difference does not necessarily mean that these topics are less aligned with the journal. It

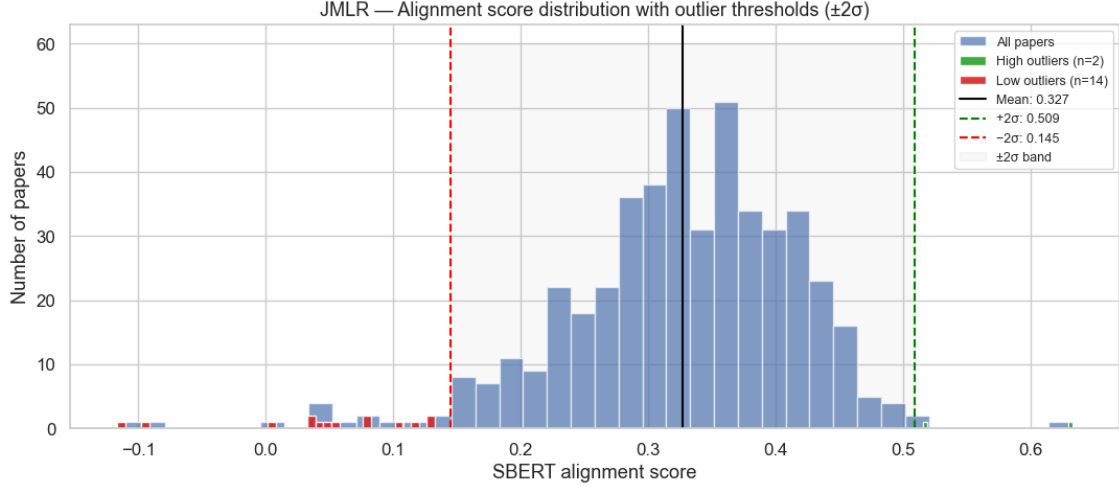


Figure 4: SBERT alignment score distribution with 2σ outlier thresholds. High outliers ($n=2$) are shown in green, low outliers ($n=14$) in red.

Table 5: BERTopic discovered topics

Topic ID	Label	Papers
-1	-1_performance_data_problem_novel	97
0	0_causal_effect_effects_identifiability	22
1	1_time_series_series_dynamics_innovations	10
2	2_networks_neural networks_neural_relu	39
3	3_instance_lstm_vision_scaling	13
4	4_gans_generative_evaluation_metrics	16
5	5_graph_graphs_clustering_embedding	41
6	6_tensor_lowrank_matrix_pca	19
7	7_lasso_regression_penalty_sparse	20
8	8_posterior_variational_inference_bayesian	51
9	9_loss_classification_training_risk	60
10	10_sampling_mala_langevin_carlo	13
11	11_policy_reinforcement_learning_reinforcement...	29
12	12_stochastic_convergence_optimization_convex	37

mostly shows a limitation of using the Aims & Scope text as the reference point, because one short reference document cannot fully cover every subfield that the journal may legitimately publish.

The noise class contains 97 papers, which is around 21% of the corpus, but this should not be treated as a failure of the model. When the clustering was forced by lowering `min_topic_size`, the results became poor and were dominated by very generic terms. This suggests that these papers are not just missed by the model. Rather, many of them are methodologically unique or interdisciplinary papers that do not fit clearly into one clean topic category.

3.7 Topic Drift Over Time

To add a temporal view of the corpus, the topic-year heatmap and topic drift visualization were used. The heatmap in Figure 6 reveals topic-level stability and isolated anomalies while Figure 7 shows the publication share of the six largest topics over time.

Table 6: Topic alignment summary sorted by mean SBERT alignment score.

Topic Label	Papers	Mean Alignment	Std
posterior, variational, inference	51	0.372	0.092
networks, neural networks, neural	39	0.364	0.047
loss, classification, training	60	0.355	0.087
policy, reinforcement learning, reinforcement	29	0.355	0.061
stochastic, convergence, optimization	37	0.348	0.072
gans, generative, evaluation	16	0.306	0.055
sampling, mala, langevin	13	0.303	0.062
tensor, lowrank, matrix	19	0.303	0.052
lasso, regression, penalty	20	0.290	0.050
graph, graphs, clustering	41	0.279	0.110
instance, lstm, vision	13	0.274	0.083
time series, series, dynamics	10	0.253	0.145
causal, effect, effects	22	0.250	0.085

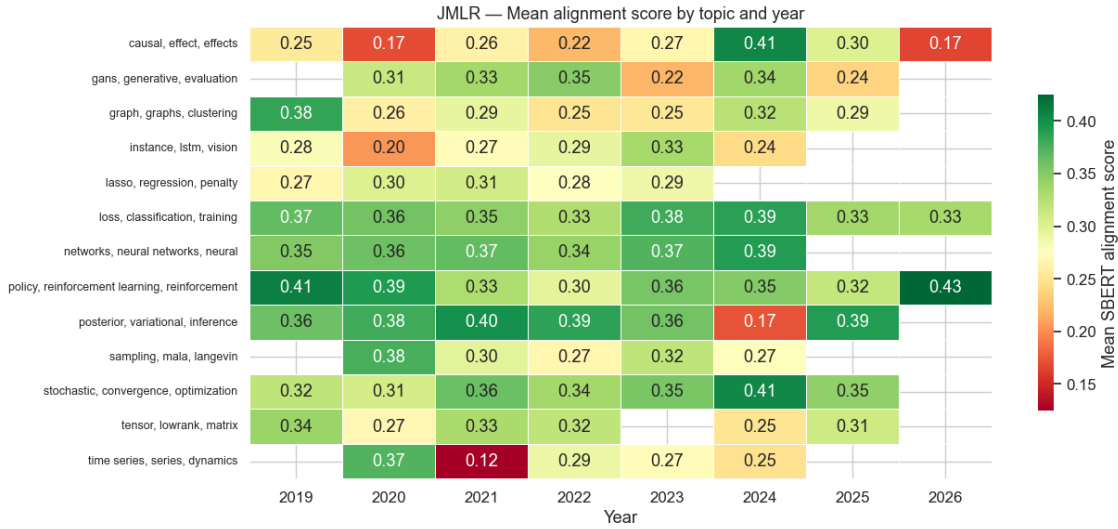


Figure 6: Mean SBERT alignment score by topic and year. Empty cells indicate years with no papers assigned to that topic.

The heatmap shows that most topics are quite stable over time, with only a few unusual cases. In general, the alignment scores stay consistent across the years, mostly between 0.25 and 0.40. This suggests that the thematic structure of the journal does not change dramatically from year to year.

The clearest exception is the time series, series, dynamics topic. In 2021, this topic had a mean score of only 0.12, which is the lowest value in the whole heatmap. This low score is probably caused by the small size of this topic, since it includes only 10 papers across all years. So, the 2021 subset seems to contain papers that were unusually far from the Aims & Scope text.

A similar drop can be seen for the posterior, variational, inference topic. Its score falls sharply to 0.17 in 2024, but then rises again to 0.39 in 2025. This suggests that the drop is more likely a temporary topic-year anomaly rather than a permanent shift in the journal’s scope.

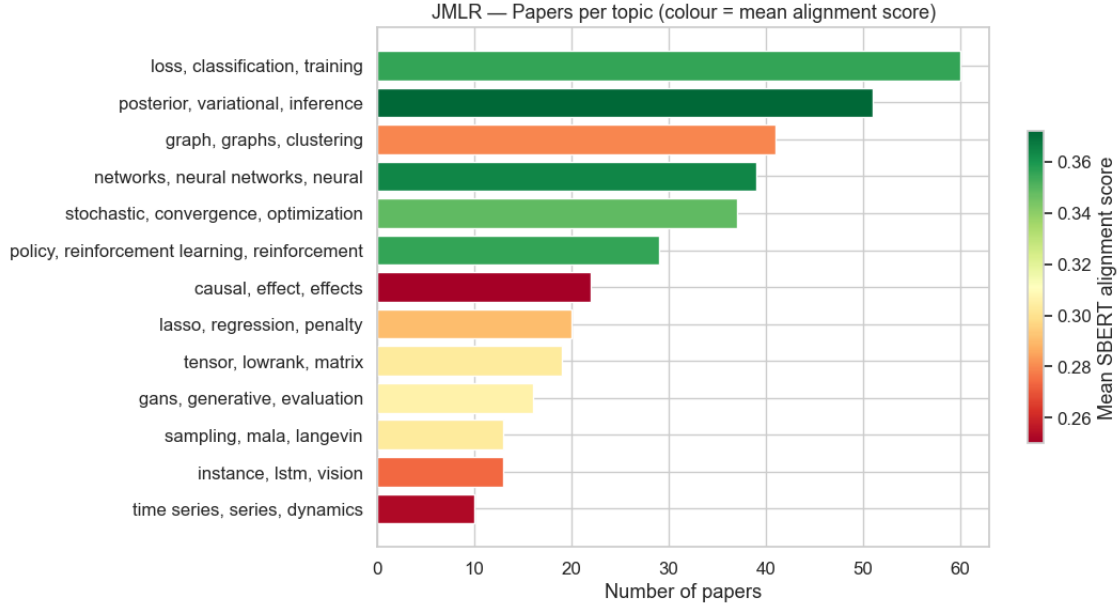


Figure 5: Number of papers per topic, colored by mean SBERT alignment score.

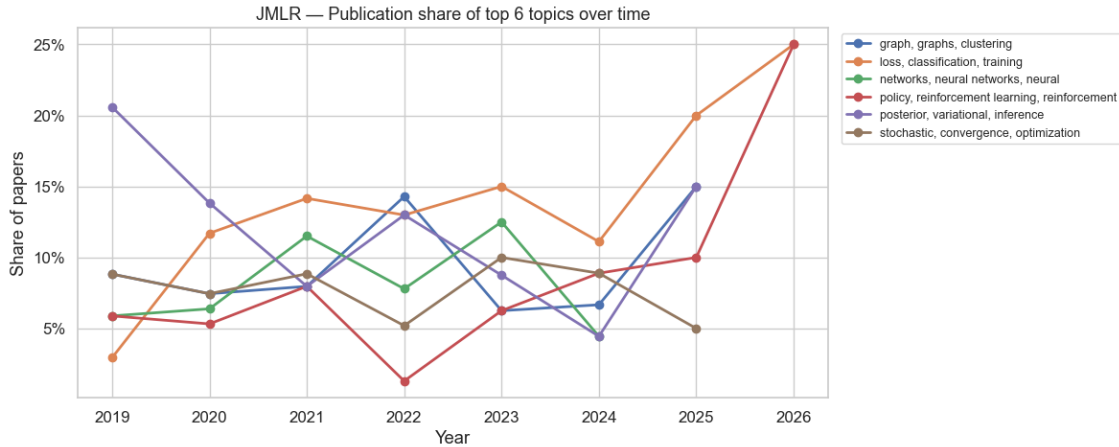


Figure 7: Share of publications belonging to the six largest topics per year (2019–2026).

Several clear patterns can be seen in Figure 7. The *posterior, variational, inference* topic was the main cluster in 2019, making up more than 20% of that year’s papers. However, its share gradually decreased and reached below 5% by 2024, before showing a partial recovery later.

In contrast, the *loss, classification, training* topic shows the opposite trend. It is the largest topic overall, with 60 papers, and its share increased from around 3% in 2019 to 25% in 2026. This makes it the clearly dominant topic in the most recent years. The *policy, reinforcement learning, reinforcement* topic also jumps to 25% in 2026, although this should be interpreted carefully because the small number of papers in 2026 can make percentages look larger than they really are.

The *graph, graphs, clustering* topic was also important in 2019 and 2022, but its share has declined since then. Overall, these changes suggest that JMLR’s published content in this corpus has gradually shifted more toward classification, optimization, and deep learning-related work, while moving away from the stronger probabilistic and Bayesian focus that appeared more clearly

in the earlier years.

3.8 Embedding Space Visualization

To complement the numerical alignment scores and topic-level results, UMAP was used to project the SBERT abstract embeddings into a two-dimensional space. This visualization helps inspect whether papers with similar semantic content are located close to one another and whether alignment patterns are visible in the embedding structure. Since UMAP preserves local neighborhood relationships, it is useful for visually exploring clusters, thematic regions, and possible outliers in the corpus.

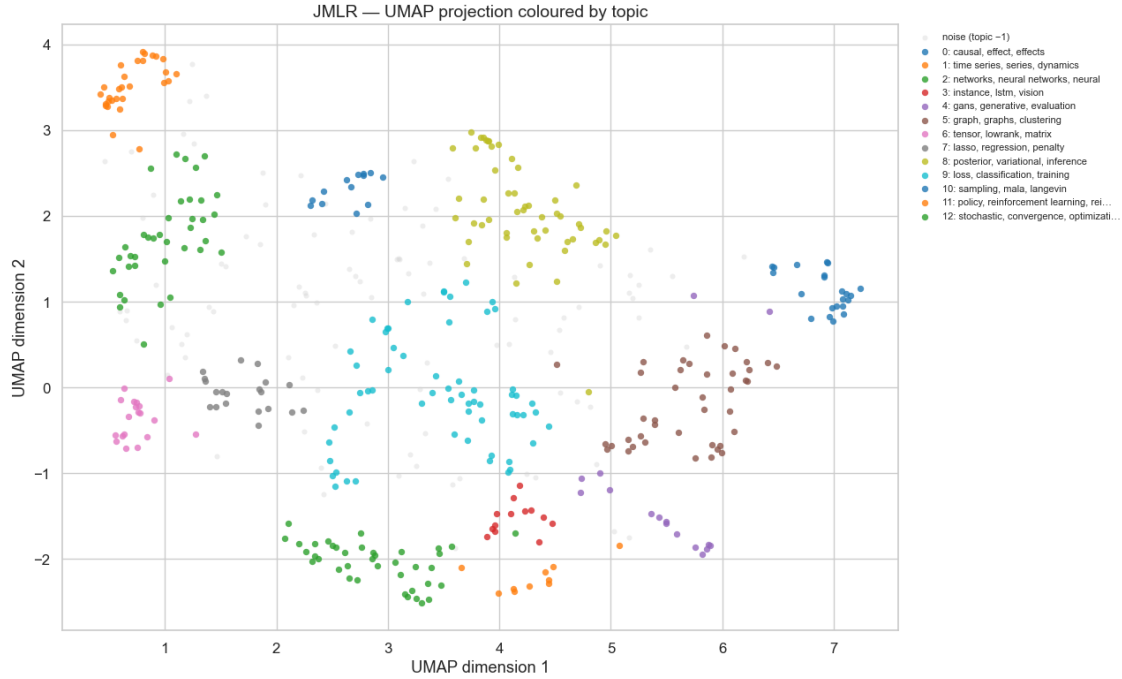


Figure 8: UMAP projection colored by BERTopic topic assignment. Noise papers (topic -1) are shown in light gray.

The topic-colored projection shown in Figure 8, confirms that BERTopic is not assigning topics randomly. The clusters also make sense geometrically in the SBERT embedding space, meaning that papers with similar themes are actually located close to each other.

Several topic groups appear clearly separated. For example, the reinforcement learning cluster is visible in orange in the upper-left area, the causal inference cluster appears in dark blue on the far left, and the loss/classification cluster is shown in cyan or teal near the center. These clear boundaries suggest that the topic labels reflect real semantic structure in the corpus.

The noise papers are spread around the edges of the space instead of forming one clear cluster. This supports the earlier interpretation that they are not one hidden topic missed by the model. Rather, they are methodologically diverse papers that do not fit neatly into a single thematic group.

The alignment-colored projection, Figure 9, gives another useful view of the results. High-scoring papers, shown in green, are spread across the central and upper parts of the space. In contrast, lower-scoring papers, shown in red or orange, are mostly located around the periphery and the left edge of the projection.

The low outliers, marked with red circles, follow the same pattern. They appear close to the extreme edges of the projection, which supports the numerical results from the cosine similarity

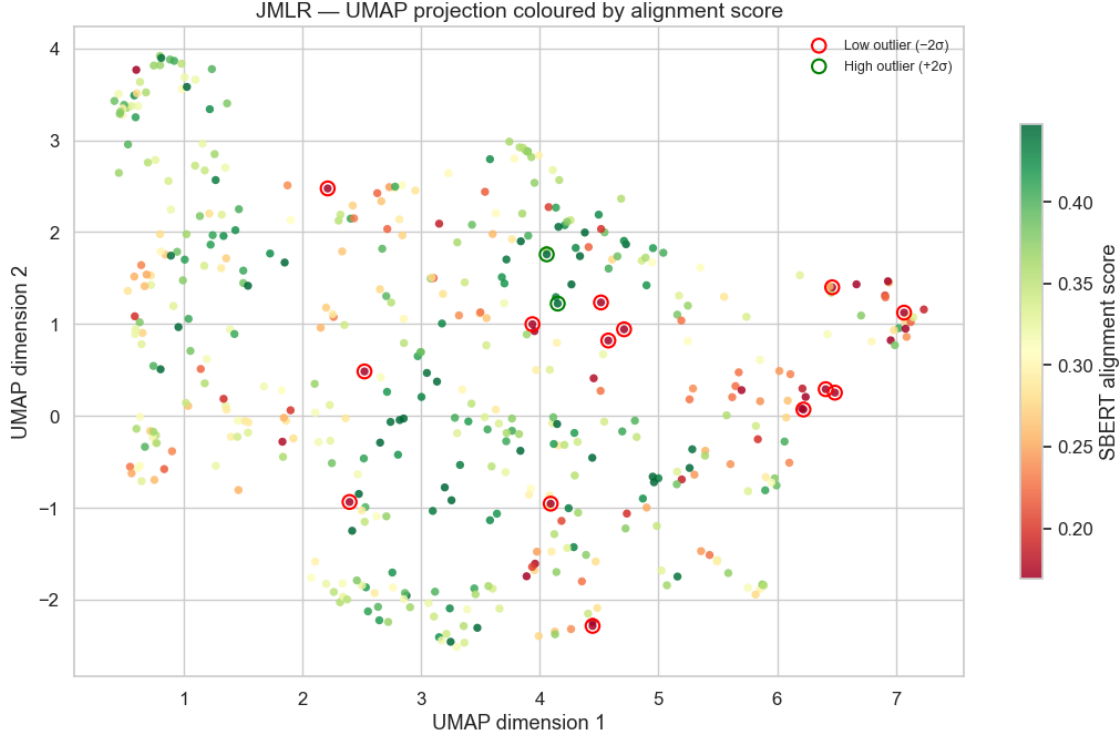


Figure 9: UMAP projection colored by SBERT alignment score (green = high, red = low). Outliers are circled.

scores. In other words, these papers are not only low-scoring numerically, but also geometrically distant from the main body of JMLR-like content.

The two high outliers, shown with green circles, are located inside the dense central region. This confirms that they are genuinely core machine learning papers, not strange isolated cases with artificially high scores.

The year-colored projection in Figure 10 shows that papers from different years are highly mixed across the semantic space. There is no clear separation by publication year. Older papers, such as 2019 papers shown in dark blue, and newer papers, such as 2025–2026 papers shown in red or orange, appear in the same regions of the projection.

This supports the result from Section 3.3, where the temporal drift slope was close to zero. In other words, the main semantic areas that define JMLR content seem to have remained stable over time. The journal has not moved into a completely different thematic space during the period covered by the corpus.

There are only small visible differences. For example, some 2019 papers are slightly more concentrated in the upper-left cluster, which corresponds to the reinforcement learning and causal inference region. Some recent 2026 papers are also more visible in the central-right and upper parts of the projection. However, these patterns are not strong enough to suggest systematic drift. They are more likely due to sampling variation.

4 Concluding Remarks

This project investigated whether articles published in the Journal of Machine Learning Research are thematically aligned with the journal’s stated Aims & Scope. Using a corpus of 467 article abstracts from 2019 to 2026, the study represented both the journal scope and article abstracts using TF-IDF and SBERT embeddings, then computed cosine similarity scores to quantify the-

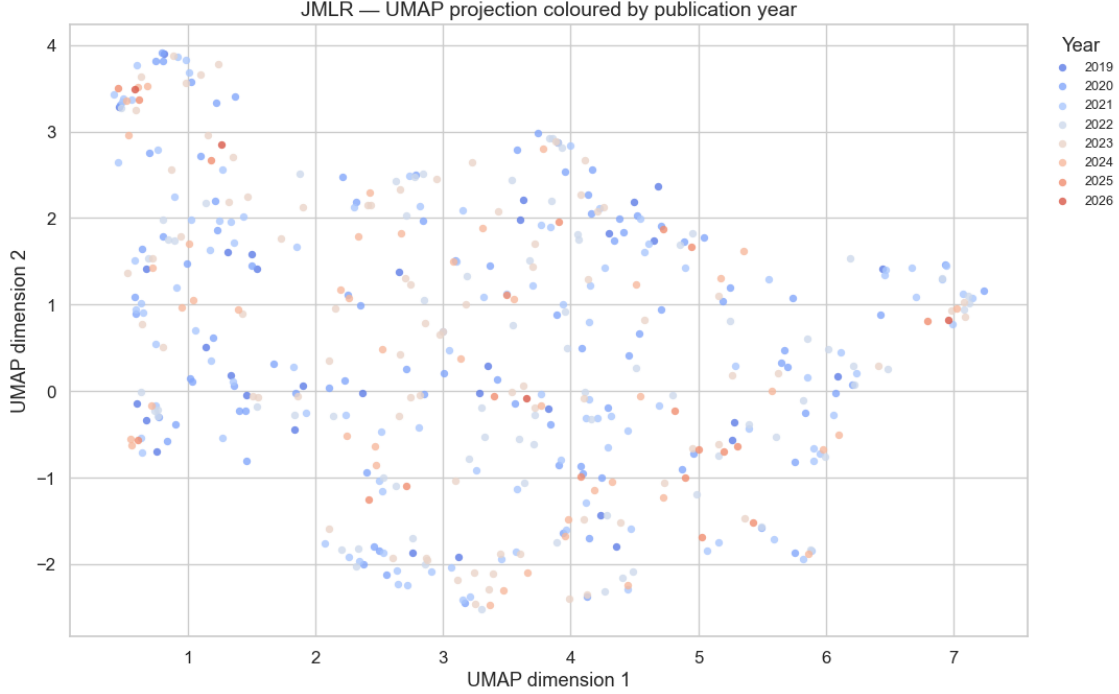


Figure 10: UMAP projection colored by publication year (blue = 2019, red = 2026).

matic alignment. The SBERT-based scores were used as the main metric because they capture semantic similarity beyond direct vocabulary overlap.

The results suggest that JMLR maintains a broadly stable thematic relationship with its stated scope. The average SBERT alignment score was approximately 0.327, indicating moderate semantic alignment across a diverse machine learning corpus. The temporal drift analysis showed only a very small negative trend of around -0.0007 per year, which does not provide strong evidence of major thematic drift during the analyzed period. Topic modeling and UMAP visualizations further supported this interpretation: the corpus contains multiple thematic clusters, but papers from different years remain broadly mixed in the embedding space rather than forming clearly separated chronological groups.

The outlier analysis identified 14 low-alignment papers and 2 high-alignment papers using the 2σ threshold. These outliers are useful for qualitative inspection, but they should not be interpreted too aggressively. A low alignment score does not necessarily mean that a paper is outside the journal’s scope; it may reflect specialized terminology, interdisciplinary framing, abstract-level ambiguity, or metadata noise from the collection process. Similarly, high-alignment papers often use language that is more directly connected to central machine learning themes and therefore naturally score higher.

The project has several limitations. First, the analysis is based only on abstracts, not full papers. Abstracts summarize the main contribution but may omit technical details that would better reveal the paper’s relevance to the journal. Second, the Aims & Scope statement is treated as a single static reference text, even though the field of machine learning and the journal’s editorial priorities may evolve over time. Third, embedding-based similarity provides a useful semantic signal, but it is not a perfect substitute for expert editorial judgment. Finally, the dataset was collected through metadata records, so possible indexing or retrieval noise may affect some results, especially among low-alignment outliers.

Future work could extend this analysis in several directions. A stronger version of the study could use full-text papers instead of abstracts, compare multiple journals within machine learning, or model the Aims & Scope dynamically over time. Another useful extension would be to

combine semantic similarity with citation networks, keywords, author-provided subject categories, or reviewer/editorial metadata. More advanced drift detection methods could also be applied to identify whether specific subfields are becoming more or less central to the journal. Overall, the project shows that embedding-based analysis can provide a practical and interpretable framework for studying thematic coherence, outliers, and conceptual evolution in scientific publishing.

source code:<https://github.com/mohmmadhadi/jmlr-alignment>

Generative AI was used as an auxiliary tool during the development and writing of this project. The main model used was Claude Sonnet 4.6. It was used to polish code, improve the clarity and structure of written text, and assist in converting the original notebook implementation into an object-oriented repository structure. The project design, methodological choices, experimental execution, and interpretation of results were carried out by the author. AI-generated outputs were manually reviewed, edited, and verified before being incorporated into the final submission.